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Research paper

Spatiotemporal violence detection in surveillance video via CNN-ConvLSTM with temporal attention fusion

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ABSTRACT

Video anomaly detection in surveillance video is a critical task in computer vision for public safety. With the exponential growth of video data and the rising occurrence of violent incidents, reliance on manual monitoring is inefficient and time-consuming. Various approaches, ranging from handcrafted methods to deep learning models, have been developed for automated violence detection. However, many existing techniques still exhibit significant limitations in achieving high accuracy, strong generalization, and real-time responsiveness. In this paper, we propose a deep hybrid framework designed for violence detection. The proposed model combines a custom lightweight convolutional neural network (CNN) for spatial feature extraction, a convolutional long short-term memory (ConvLSTM) layer to capture spatiotemporal dependencies, and a temporal attention module to focus on the most informative motion patterns. The proposed architecture effectively captures appearance and motion features, achieving high performance with low computational cost while ensuring consistent performance across multiple datasets, making it suitable for real-world surveillance applications. Comprehensive experiments on four benchmark datasets (Hockey Fights, RLVS, Surveillance Fight, and AIRTLab) demonstrate strong performance, achieving accuracies up to 99.15% and F1-scores up to 99.37%. Compared to pretrained models (VGG16, VGG19, MobileNetV2), the custom CNN-based approach achieves improvements of up to 2.24% in accuracy, 2.22% in F1-score, and 0.35% in AUC, while reducing inference time by 12% to 56.9%. Additionally, compared to 19 state-of-the-art methods, the proposed model achieves accuracy improvements ranging from 0.24% to 30.7%, confirming its effectiveness for real-world violence detection applications.

1. Introduction

In recent years, the increasing rate of crime and violent activities has posed serious threats to public safety, making surveillance systems essential for monitoring environments such as educational institutions, commercial spaces, and transportation hubs. However, these systems generate vast amounts of video data, while violent events occur relatively infrequently. As a result, manual monitoring becomes inefficient, time-consuming, and prone to human error. Additionally, the large volume of redundant video data imposes significant storage demands. These challenges highlight the need for automated and intelligent systems capable of accurately detecting violent or anomalous activities while reducing human effort and storage requirements [1–3].

Recent advances in artificial intelligence (AI), particularly deep learning (DL), have significantly improved the performance of video surveillance systems. Unlike traditional methods that required handcrafted features [4–6], DL models, especially convolutional neural networks (CNNs), can automatically learn and extract important patterns directly from raw data [7–9]. However, CNNs primarily focus on spatial feature extraction and lack the ability to capture temporal dependencies across video frames. To address these limitations, recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM) [10], Bidirectional LSTM (Bi-LSTM) [11,12], and Gated Recurrent Units (GRUs) [13] are widely used to model long-term dependencies and capture contextual information from both past and future frames.

Despite their effectiveness, these approaches often require flattening

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spatial features before temporal processing, leading to the loss of important spatial information and limiting the model's ability to preserve spatiotemporal structures. To overcome this limitation, 3D CNN-based approaches have been introduced to jointly capture spatial and temporal features. However, these models require high computational resources, large memory, and long training times, limiting their suitability for real-time surveillance applications and deployment in resource-constrained environments.

More recently, attention mechanisms have been introduced to focus on the most informative regions in video data [14,15]. By dynamically assigning different weights to input features, attention mechanisms enable models to concentrate on the most relevant information in images or videos. Several types of attention mechanisms, such as spatial, temporal, channel-wise, and spatiotemporal attention, have been effectively applied to enhance the understanding of complex videos [16–18]. In recent years, hybrid frameworks combining CNNs, recurrent networks, attention mechanisms, and transformers [19,20] have shown promising performance in violence detection tasks. However, many of these models employ complex multi-stage architectures that increase computational demands and compromise real-time deployment. Furthermore, transformer-based and attention-heavy designs introduce high memory consumption and longer inference time. In addition, some methods depend on multimodal inputs or multiple processing modules, reducing their practicality in real-world surveillance scenarios.

To address these limitations, this work proposes a lightweight and unified hybrid framework that combines a custom CNN for efficient spatial feature extraction, ConvLSTM to preserve spatiotemporal structures without flattening, and a temporal attention module to focus on the most informative frames. This design achieves a balance between accuracy and computational efficiency while ensuring consistent performance across multiple datasets, making it suitable for real-time surveillance applications. The key contributions of this research are summarized as follows:

- A custom CNN architecture is proposed, consisting of six convolutional blocks followed by a dropout layer. The proposed lightweight design efficiently captures spatial features from surveillance video frames, reducing processing time without sacrificing accuracy in violent action recognition.
- ConvLSTM is employed to capture spatiotemporal dependencies while preserving spatial information across video sequences.
- A temporal attention module is incorporated to guide the model's focus toward the most relevant segments of the video, enhancing both the accuracy and responsiveness of early violence detection.
- Pre-trained models (VGG16, VGG19, and MobileNetV2) are used as alternative feature extractors, combined with ConvLSTM and Temporal Attention to evaluate their effectiveness in violence detection compared to the proposed CNN.
- The hybrid models are evaluated on four publicly available datasets (Hockey Fights, RLVS, Surveillance Fight, and AIRTLab), which are related to fighting or violent action.
- A comparative study is conducted in the conventional approaches for violence detection in surveillance videos to assess the effectiveness of the proposed method.

The organization of this paper is as follows: Section 2 reviews the related work of violence detection in surveillance videos, Section 3 details the proposed framework, Section 4 presents the datasets used in this study, evaluation metrics, and experimental results. Finally, Section 5 concludes the paper and outlines directions for future work.

2. Related works

Violence detection in surveillance videos is an important research area aimed at enhancing public safety and enabling timely threat response. Various computational approaches have been developed to detect violent and abnormal behaviors while reducing reliance on human monitoring. These approaches are categorized into Conventional Handcrafted Feature-Based Methods, Spatial–Temporal Deep Learning Models, and Attention-Based Hybrid Models, which are discussed in detail in the following subsections.

2.1. *Conventional handcrafted feature-based methods*

Early violence detection systems relied primarily on traditional machine learning techniques with manually designed feature extractors to capture motion patterns and visual cues from video footage. Khalaf et al. [4] proposed a violence detection framework that utilized Principal Component Analysis (PCA) for feature extraction followed by multiple machine learning classifiers, including Logistic Regression, Naive Bayes, AdaBoost, and Stochastic Gradient Descent to distinguish between violent and non-violent actions. Similarly, Jayasree et al. [5] introduced a cascaded approach combining motion-based frame selection with handcrafted features such as Scale-Invariant Feature Transform (SIFT), Histogram of Optical Flow (HoF), and Motion Boundary Histogram (MBH), which are fused into the motion boundary SIFT (MoBSIFT) descriptor for classification. Other studies focused on extracting spatiotemporal or statistical features, such as Deepak et al. [6], who used motion descriptors with an support vector machine (SVM) classifier, and Wintarti et al. [21], who employed wavelet-based features with SVM. Likewise, Lohithashva et al. [22] utilized the Local Optimal Oriented Pattern (LOOP) descriptor combined with SVM, while Jaiswal et al. [23] introduced an ensemble boosting method based on low-level feature extraction. Although these approaches are computationally efficient and relatively simple to implement, they often struggle in complex, real-world scenarios. Their performance degrades under low-quality or diverse conditions, and their reliance on handcrafted features limits their ability to capture subtle and complex patterns of violent behavior. Consequently, these limitations have driven the shift toward more advanced automated feature learning techniques to enhance robustness and accuracy.

2.2. *Spatial–Temporal deep learning models*

The advent of deep learning marked a significant paradigm shift in violence detection, enabling automated learning of hierarchical spatiotemporal representations without manual feature engineering. Initially, early approaches focused on spatial feature extraction using pretrained CNNs, such as Jebur et al. [7], who fused features from Xception, InceptionV3, and InceptionResNet with ML classifiers, though lacking temporal modeling and showing limited generalization across datasets. To address this, many studies adopted CNN–RNN hybrid architectures to jointly model spatial and temporal information. Vosta et al. [24] combined ResNet50 with ConvLSTM for crime anomaly detection, while Vijeikis et al. [25] used U-Net with a MobileNetV2 encoder and LSTM for violence classification. Similarly, Haque et al. [26] proposed BrutNet, which used CNN for spatial feature extraction and GRU for temporal modeling, followed by six dense layers for classification. Moreover, Asad et al. [27] extracted spatial features using VGG-16 with residual blocks followed by LSTM and Sharma et al. [28] employed Xception with LSTM for sequence prediction. In addition, Contardo et al. [29] combined MobileNetV2 with recurrent layers. Furthermore, ensemble strategies were explored by Kaur et al. [10], using multiple CNNs with stacked LSTM. Natha et al. [30] proposed a Stack Ensemble Road Anomaly Detection (SERAD) model, which is stacked ensemble of pretrained CNNs (VGG19, ResNet50, and InceptionV3) that improves anomaly detection accuracy over individual models on road surveillance datasets. Additionally, Vrskova et al. [31] utilized time-distributed Conv2D with ConvLSTM to learn spatiotemporal patterns directly from video sequences.

In parallel, 3D CNN-based methods were developed to directly capture spatiotemporal features. Dündar et al. [9] proposed a simple 3D

CNN with two convolution layers but faced high computational cost. Similarly, Magdy et al. [32] introduced a 4D model combining RGB frames and optical flow processed by 3D ResNet-based residual blocks. Moreover, Iftee et al. [11] proposed VioNET, combining a 3D CNN and Vision Transformer for spatial feature extraction with a Bi-LSTM for temporal modeling. Sernani et al. [8] used pretrained C3D for feature extraction followed by SVM or dense layers for classification. More efficient designs include DeepGuard by Provath et al. [33], which employs MoViNet-A0 with keyframe selection. Baca et al. [34] proposed a real-time violence recognition framework composed of three main modules: a Spatial Motion Extractor (SME) to identify regions of interest, a Short Temporal Extractor (STE) based on MobileNetV2 to capture motion features, and a Global Temporal Extractor (GTE) to model long-term temporal dependencies and refine the overall representation.

Overall, although DL-based methods enhance detection performance through rich spatiotemporal representations, challenges such as computational complexity, accuracy variations across datasets, and real-time applicability still remain.

2.3. Hybrid attention-based models

To address the limitations of conventional deep learning architectures, recent research has incorporated attention mechanisms that enable models to selectively focus on discriminative spatial regions and critical temporal segments. Several studies integrate attention mechanisms within CNN–RNN frameworks. For instance, Deshpande et al. [35] combined residual attention with ConvLSTM for abnormal activity recognition. Vosta et al. [14] proposed KianNet, which integrates ResNet50 and ConvLSTM with Multi-Head Self-Attention (MHSA) to enhance and refine spatiotemporal feature representation. Similarly, Ullah et al. [36] and Khalfaoui et al. [37] employed lightweight CNNs (MobileNet variants) with attention-enhanced LSTM/Bi-LSTM for efficient spatiotemporal modeling. Wang et al. [15] introduced Bi-Directional Long-Term Motion Attention (Bi-LTMA) with EfficientNet-B0 and temporal modules to better capture motion dynamics. In addition, transformer-based and hybrid attention models have been explored to capture long-range dependencies. Rendón-Segador et al. [17] proposed ViolenceNet, which combines 3D DenseNet with Multi-Head Self-Attention (MHSA) and Bidirectional ConvLSTM for spatiotemporal feature learning. Moreover, Natha et al. [20] proposed a CNN–BiLSTM framework enhanced with transformer-based attention for modeling short- and long-term dependencies, while Ahmed et al. [38] introduced a multimodal transformer framework integrating visual, audio, and motion features. Additionally, Khan et al. [19] proposed VD-Net, which utilizes temporal convolutional networks with bottleneck attention and transformer modules to enhance feature representation. Dawoud et al. [39] proposed FusedVision, a fusion-based framework that combines a normalcy learning module based on autoencoders, a detection module for object localization, and an attention-driven merging module to effectively integrate multimodal information for accurate anomaly detection.

3. Proposed model

This section introduces the proposed model architecture for detecting violence in surveillance videos. This approach improves recognition accuracy while maintaining consistent performance across datasets and low latency for real-time applications. The model integrates the strengths of CNNs, ConvLSTM networks, and a Temporal Attention mechanism. The process begins with a custom CNN that extracts spatial features from individual video frames. These features are then passed into a ConvLSTM network, which captures how actions develop over time while preserving spatial context. Finally, a Temporal Attention Module helps the model focus on the most important moments in the video, allowing it to better understand and identify violent behavior. This combination creates an accurate system capable of handling the complexities of real-world surveillance footage. Fig. 1 presents the overall architecture of the proposed Violence Detection System, and the details of each module are described in the following subsections.

3.1. Preprocessing module

Preprocessing is the initial step applied before building the model. Each video was divided into 16 consecutive frames (chunks) to form fixed-length temporal sequences, ensuring a uniform representation across all samples. To handle the varied resolutions of the video clips, all frames are resized to 224×224 pixels, ensuring a consistent input size across the dataset.

3.2. Spatial features extraction module

In this module, spatial features from the individual video frames are extracted based on a custom lightweight CNN. The proposed lightweight CNN architecture consists of six convolutional blocks, where each block includes a convolutional layer followed by batch normalization and max-pooling. The use of six convolutional blocks provides a balance between model complexity and computational efficiency. A shallow architecture may fail to capture complex spatial patterns, while a deeper network significantly increases computational cost and may lead to overfitting, particularly with limited training data. Therefore, a moderate-depth design is adopted to ensure effective feature extraction while maintaining efficiency suitable for real-time applications. Table 1 shows the proposed custom CNN architecture, where the convolutional layers progressively increase from 16 to 512 filters across the network layers. This design enables the model to learn hierarchical feature representations, starting from low-level features such as edges and textures in early layers to more complex and discriminative patterns in deeper layers. Each convolutional layer uses a 3×3 kernel with a stride of 1×1, which is a standard choice in modern CNN architectures due to its ability to effectively capture local spatial features while keeping the number of parameters low. The ReLU activation function is applied after each convolutional layer to introduce non-linearity, enabling the model to learn complex patterns. Following each convolution, batch normalization is performed to stabilize and accelerate training, while max-pooling with a 2×2 kernel and a stride of 2×2 reduces the spatial dimensions, helping the network focus on the most important features. Additionally, a dropout layer is included at the end of the model to reduce overfitting and enhance generalization.

3.3. Spatiotemporal feature extraction module

Traditional models typically use CNNs to extract spatial features and LSTMs to capture temporal information. However, this approach often results in the loss of valuable spatial relationships, as the feature maps are flattened before being processed by the LSTM. To address this limitation, ConvLSTM [40] integrates convolution operations within the LSTM structure, allowing the model to preserve spatial structures while simultaneously learning temporal dynamics across time steps. This integration makes ConvLSTM particularly well-suited for video analysis tasks, as it can effectively learn spatiotemporal patterns. By combining CNNs for spatial feature extraction with ConvLSTM for spatiotemporal modeling, the system benefits from retaining spatial information throughout the sequence, leading to improved performance.

In the proposed framework, the ConvLSTM layer operates on spatial feature maps of size (B, T, H, W, C), where B denotes the batch size, $T = 16$ corresponds to the number of input frames, H, W, and C represent the spatial height, width, and number of channels. As illustrated in Fig. 1, the layer receives an input of shape (B, 16, 3, 512). It is configured with 64 filters and a kernel size of 3×3, producing an output tensor of (B, 16, 1, 1, 64). This design enables the model to effectively capture both spatial and temporal dependencies across the sequence. The use of 64 filters reflects a practical trade-off, offering sufficient representational

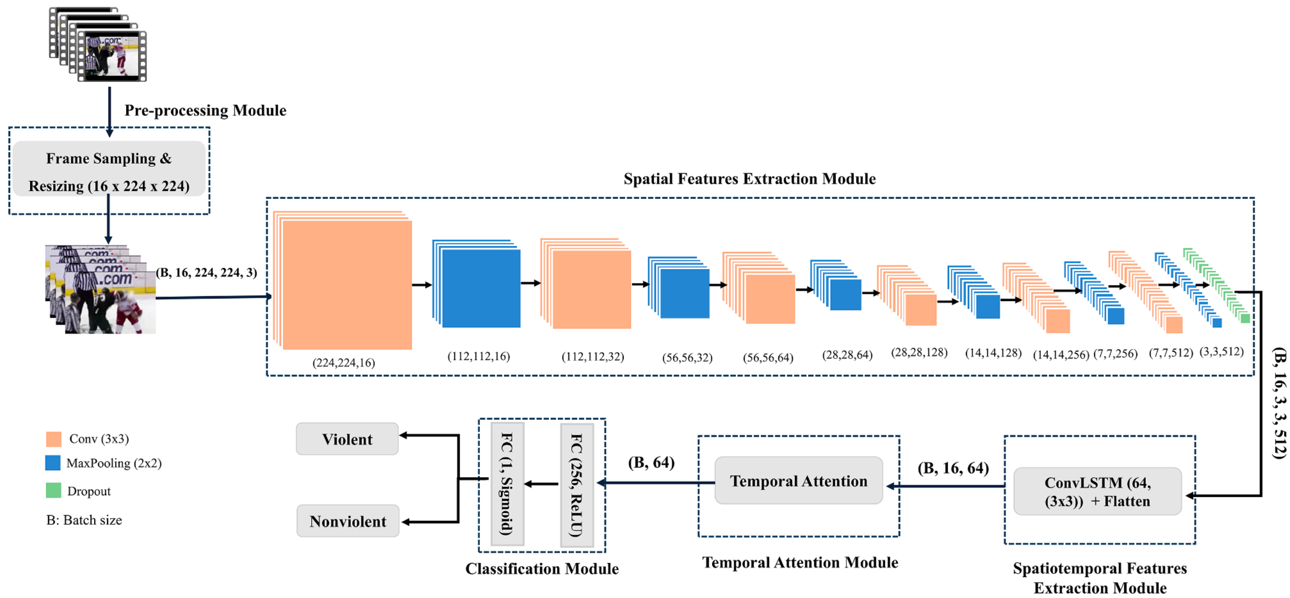

Fig. 1. Overall architecture of the proposed Violence Detection System.

Table 1
Detailed Architecture of the Proposed CNN Model, Including Layer Configurations, Kernel Sizes, and Output Shapes.

Layers	Stride	No. of Kernels	Kernel_Size	Output_shape
conv2d_1	1 × 1	16	3 × 3	(B, 224, 224, 16)
batch_normalization_1	-	-	-	(B, 224, 224, 16)
max_pooling2d_1	1 × 1	-	2 × 2	(B, 112, 112, 16)
conv2d_2	1 × 1	32	3 × 3	(B, 112, 112, 32)
batch_normalization_2	-	-	-	(B, 112, 112, 32)
max_pooling2d_2	1 × 1	-	2 × 2	(B, 56, 56, 32)
conv2d_3	1 × 1	64	3 × 3	(B, 56, 56, 64)
batch_normalization_3	-	-	-	(B, 56, 56, 64)
max_pooling2d_3	1 × 1	-	2 × 2	(B, 28, 28, 64)
conv2d_4	1 × 1	128	3 × 3	(B, 28, 28, 128)
batch_normalization_4	-	-	-	(B, 28, 28, 128)
max_pooling2d_4	1 × 1	-	2 × 2	(B, 14, 14, 128)
conv2d_5	1 × 1	256	3 × 3	(B, 14, 14, 256)
batch_normalization_5	-	-	-	(B, 14, 14, 256)
max_pooling2d_5	1 × 1	-	2 × 2	(B, 7, 7, 256)
conv2d_6	1 × 1	512	3 × 3	(B, 7, 7, 512)
batch_normalization_6	-	-	-	(B, 7, 7, 512)
max_pooling2d_6	1 × 1	-	2 × 2	(B, 3, 3, 512)
Dropout	-	-	-	(B, 3, 3, 512)

Total params: 1576800.
Trainable params: 1574784.
Non-trainable params: 2016.
B denotes the batch size.

capacity while maintaining computational efficiency and reducing the dimensionality of the feature maps.

Building on this design, a deeper understanding of the ConvLSTM layer can be gained by examining its internal mechanism. ConvLSTM extends the traditional LSTM by replacing matrix multiplications with convolutional operations, making it more suitable for spatiotemporal data. Similar to the standard LSTM, it maintains memory cells regulated by three gates, forget, input, and output [40], as illustrated in Fig. 2. In this structure, x_t refers to the input at time t, C_{t-1} and C_t denote the previous and current cell states, and h_{t-1} and h_t represent the past and current hidden states, respectively. The forget (f_t), input (i_t), and output (o_t) gates regulate the flow of information, enabling the model to store and retrieve relevant features over time. Each gate in ConvLSTM operates on 3D tensors (e.g., height × width × channels) instead of 1D vectors, allowing it to capture spatial and temporal information in the data. The operations of ConvLSTM can be formally represented as shown in Eqs. (1)–(5).

$$f_t = \sigma\left(w_f * x_t + w_f * h_{t-1} + w_f \circ C_{t-1} + b_f\right) \tag{1}$$

$$i_t = \sigma(w_i * x_t + w_i * h_{t-1} + w_i \circ C_{t-1} + b_i) \tag{2}$$

$$C_t = f_t \circ C_{t-1} + i_t \circ tanh(w_C * x_t + w_C * h_{t-1} + b_C) \tag{3}$$

$$o_t = \sigma(w_o * x_t + w_o * h_{t-1} + w_o \circ C_{t-1} + b_o) \tag{4}$$

$$h_t = o_t \circ \tanh(C_t) \tag{5}$$

Where "*" refers to the convolutional operation, "∘" represents element-wise multiplication, while w's and b's represents weight and bias matrices.

3.4. Temporal attention module

A temporal attention mechanism is integrated into the CNN–ConvLSTM framework to enhance model performance and improve its ability to focus on informative temporal patterns. This module assigns adaptive importance weights to different time steps, allowing the model to emphasize frames that are more relevant to violent activities [16].

The attention module is applied to the reshaped output of the ConvLSTM layer, obtained via a TimeDistributed Flatten operation that transforms the tensor from (B, 16, 1, 1, 64) into a sequence of feature vectors of shape (B, T, F) = (B, 16, 64), where F denotes the feature dimension. A dense layer with a *tanh* activation function is then employed to compute attention scores for each time step, capturing their relative importance within the temporal sequence. These scores are normalized using the Softmax function to produce a probability distribution across the temporal sequence. A context vector is subsequently computed as a weighted sum of the feature vectors, enabling the model to focus on the most informative frames. Consequently, the resulting context vector has a shape of (B, F) = (B, 64) and provides a compact yet informative representation of the temporal features, where greater

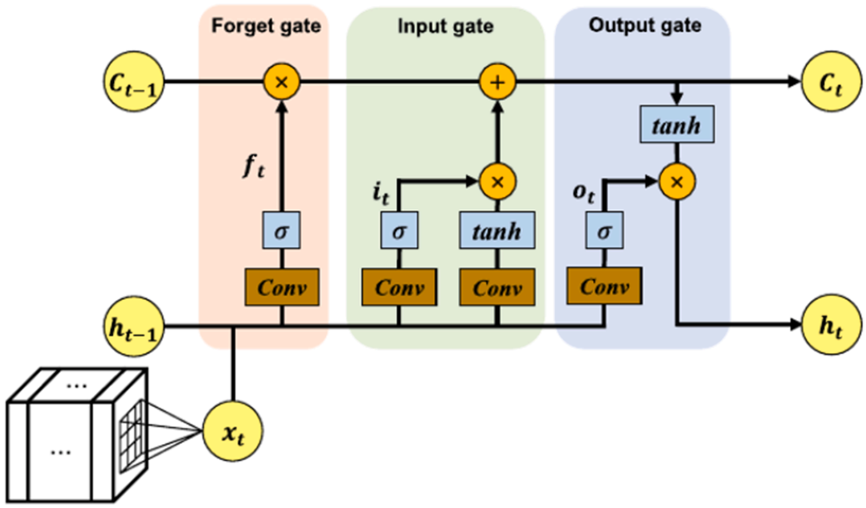

Fig. 2. Architecture of ConvLSTM [41].

emphasis is placed on the most relevant frames within the sequence. Fig. 3 presents the block diagram of the temporal attention module, while the underlying process can be mathematically formulated as follows:

$$score_t = tanh(W * v_t + b) \quad (6)$$

$$a_t = \text{SoftMax}(score_t) \quad (7)$$

$$context = \sum_{t=1}^{T} a_t . v_t \quad (8)$$

Where v_t denotes the feature vector at time t, $score_t$ refers to the attention score, a_t is the attention weights, W and b represent weight and bias matrices.

3.5. Classification module

Within this module, the spatiotemporal features extracted by the CNN, ConvLSTM, and temporal attention modules are fed into a classification stage to determine the category of the input video. The classification module comprises two fully connected layers with 256 and 1 neurons, respectively. A ReLU activation is applied in the first layer, followed by a sigmoid function in the output layer to perform binary classification (Violence vs. Non-Violence).

4. Experimental evaluation

This section provides a structured overview of the experimental process carried out to assess the performance of the proposed model. It began with a description of the experimental setup, followed by the evaluation metrics used to quantify the proposed model performance. The datasets used in the study were then introduced, highlighting their relevance and diversity. Finally, the simulation results were presented and analyzed, providing insights into the model's effectiveness across different scenarios.

4.1. Experimental setup

To evaluate the performance of the proposed violence detection model, multiple experiments were conducted on various benchmark datasets. All experiments were carried out on a workstation equipped with an Intel Xeon W-2125, 128GB RAM, and Nvidia Quadro P4000 GPU 24GB. The proposed model was implemented in Python 3.9 using the TensorFlow framework. All the video frames were resized to 224 × 224 pixels to train and test the proposed model. A cross-validation strategy was applied by randomly splitting each dataset into 80% for training and 20% for testing, while 12.5% of the training set was used as a validation set. This randomized 80 – 20 split was repeated five times to ensure reliable evaluation. The Adam optimizer was used with different learning rates (0.001, 0.0001, and 0.00001) with 8 and 16 batch sizes to explore the optimal training configuration. Additionally, an early stopping mechanism was implemented, stopping training if the model doesn't achieve improvement after 5 consecutive epochs. Furthermore, to provide a comprehensive evaluation of the implementation and computational efficiency, the hardware setup, training configuration, and runtime performance metrics are summarized in Table 2.

4.2. Evaluation metrics

Performance evaluation of the proposed model was conducted using various metrics, including accuracy, precision, recall, and F1-score, to comprehensively assess its classification performance. These metrics are calculated according to the following formulas:

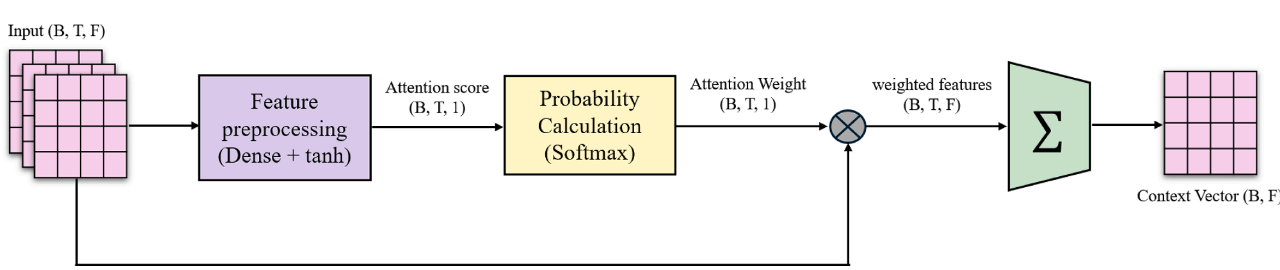

Fig. 3. Temporal Attention Module Architecture.

Table 2
Summary of implementation details, including hardware specifications, training configuration, and computational performance metrics.

Category	Parameter	Value
Hardware	CPU	Intel Xeon W-2125
	RAM	128 GB
	GPU	Nvidia Quadro P4000 (24GB)
Software	Framework	TensorFlow
	Python Version	3.9
Input Settings	Frame Resolution	224 × 224
	Batch Size	8, 16
	Optimizer	Adam
	Learning Rates	0.001, 0.0001, 0.00001
Inference	Inference Time per Frame	0.15–3.29 ms

$$Accuracy = \frac{TP + TN}{(TP + FN + FP + TN)} \tag{9}$$

$$Precision = \frac{TP}{(TP + FP)} \tag{10}$$

$$Recall = \frac{TP}{(TP + FN)} \tag{11}$$

$$F1 - score = \frac{(2 * precision * recall)}{(precision + recall)} \tag{12}$$

$$specificity = \frac{TN}{(TN + FP)} \tag{13}$$

Where True Positive (TP) denotes the number of videos that are actually positive and correctly predicted as positive, False Negative (FN) refers to positive videos incorrectly predicted as negative, False Positive (FP) represents negative videos incorrectly predicted as positive, and True Negative (TN) indicates negative videos correctly predicted as negative by the model. For these metrics, higher values indicate better model performance, as they reflect the model's ability to correctly classify instances (accuracy), precisely identify positive cases without including false positives (precision), and effectively detect all actual positive cases (recall), and correctly identify negative cases (specificity). A high F1-score indicates a good balance between precision and recall, which is particularly important in cases of class imbalance.

In addition to these metrics, the AUC metric is used to further evaluate the model's performance. AUC represents the ability of the model to distinguish between classes. It is based on the relationship between the true positive rate (recall) and the false positive rate (1 - specificity) at various classification thresholds. The AUC value ranges from 0 to 1, where high AUC values indicate the best classification performance between the positive and negative classes.

4.3. Datasets

The proposed model is evaluated on four publicly available datasets: AIRTLab, RLVS, Surveillance Fight, and Hockey Fights. These datasets cover diverse environments and a wide range of violent actions. Table 3 summarizes the dataset characteristics and splits, while Fig. 4 presents sample instances. Further details are provided in the following subsections.

4.3.1. Hockey fights dataset

The hockey fight dataset was collected in 2011 from the National Hockey League (NHL) games. It contains 1000 videos and is manually labeled as either fights or non-fights. Each category in the dataset includes 500 video clips, with a video resolution of 360 × 288 pixels and a length of 25 frames per second [42].

4.3.2. Real life violence situations (RLVS) dataset

The RLVS dataset was collected in 2019 from various environments, with video resolutions ranging from 480p to 720p It is made up of 2000 videos, split into 1000 violent and 1000 non-violent samples. The shortest video length in the dataset is 3 s, while the longest is 7 s. Video frame widths range from 224 to 1920 pixels, and heights range from 224 to 1080 pixels [43].

4.3.3. Surveillance fight dataset

The surveillance fight dataset was collected in 2020 from YouTube mostly, in addition to non-fight videos from the available surveillance datasets like synopsis and CamNet. It consists of 300 video clips, equally divided into 150 fights and 150 non-fight clips. The mean duration of the videos in the dataset is 2 s, and the videos have various frames and sizes. The dataset has different fight behaviors, such as fistfights, kicking, wrestling, and hitting with objects [44].

4.3.4. AIRTLab dataset

The AIRTLab dataset was created in 2021. It consists of 350 video clips with a resolution of 1920 × 1080 pixels. The dataset is divided into two categories: "violent", containing 230 video clips, and "non-violent", containing 120 video clips. The minimum duration video in the dataset is 2 s, while the maximum duration is 14 s. All videos are recorded at a frame rate of 30 frames per second (fps) using two cameras positioned at different locations [8].

Despite their common focus on violence detection, the datasets exhibit significant domain differences in terms of environment, visual quality, and motion complexity. The Hockey Fights dataset represents a structured setting with consistent backgrounds but involves rapid motion and player overlap. In contrast, RLVS captures highly unconstrained real-world scenarios characterized by crowd density, motion blur, and diverse backgrounds. The Surveillance Fight dataset introduces additional challenges such as low resolution, poor lighting, camera instability, and varying viewpoints typical of CCTV footage. Meanwhile, the AIRTLab dataset, although more controlled, requires the model to distinguish subtle differences between staged violent and non-violent interactions. These variations highlight the presence of domain shifts across datasets, motivating the need for robust spatiotemporal modeling.

4.4. Evaluation results

This section presents a comprehensive performance analysis of the

Table 3
Summary of dataset characteristics and splits for the Hockey Fights, RLVS, Surveillance Fight, and AIRTLab datasets.

Dataset	Violent Clips	Non-Violent Clips	Train/ chunk	Val/ chunk	Test/ chunk	Resolution	Length/ clip (S)	Environment
Hockey Fights	500	500	1404	201	402	360 × 288	1–2	Sports (ice hockey gameplay & fights)
RLVS	1000	1000	11,924	1703	3407	480 × 1080	3–7	Real-world (streets, public places, indoor scenes)
Surveillance Fight	150	150	637	92	183	Various	~2	CCTV (fight scenes)
AIRTLab	230	120	2475	354	708	1920 × 1080	2- 14	Controlled indoor

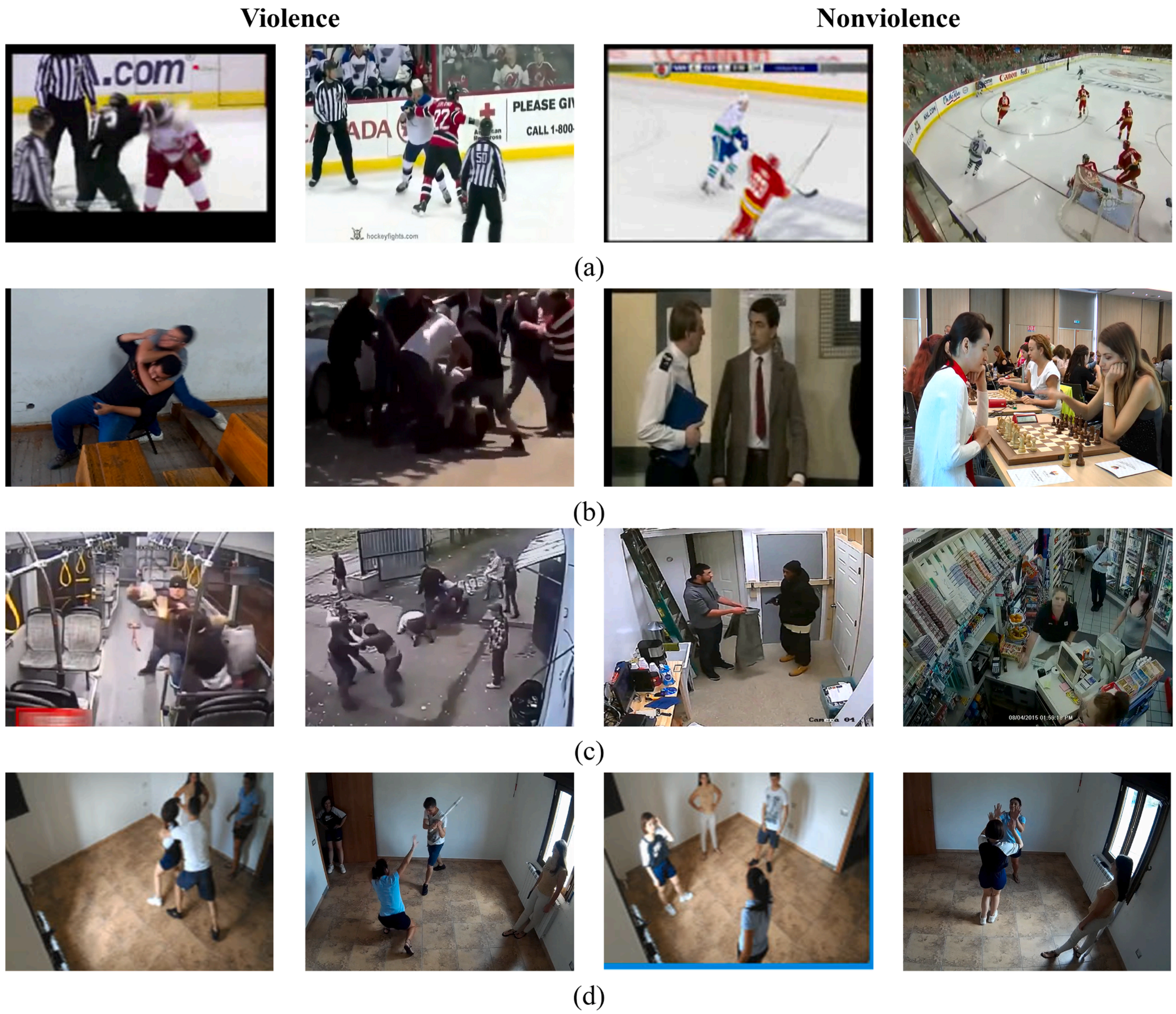

Fig. 4. Samples of datasets. (a) the Hockey Fights, (b) RLVS, (c) Surveillance Fight, and (d) AIRTLab.

proposed model from multiple perspectives. First, it was compared with variants employing pretrained CNN architectures to examine the contribution of the custom CNN design. Subsequently, the model was evaluated using both quantitative and qualitative metrics to assess its effectiveness. This was followed by a comparison with conventional CNN–LSTM-based approaches to demonstrate the advantages of the proposed framework. The detailed evaluation results are provided in the subsequent subsections.

4.4.1. Comparison with pretrained CNN-Based architectures

This subsection presents a comparative analysis between the proposed CNN + ConvLSTM + Temporal Attention model and its variants that employ pretrained CNN architectures as backbone feature extractors, namely VGG16, VGG19, and MobileNetV2. In these variants, the proposed custom CNN is replaced with each pretrained backbone while maintaining the same ConvLSTM and Temporal Attention components, ensuring a fair and controlled comparison. The optimal configuration of the proposed CNN-based model is determined through experiments on different architectural settings (number of convolutional blocks and filter sizes), while pretrained backbones retain their standard architectures; detailed results are presented in the ablation study. The objective was to evaluate the effectiveness of the proposed architecture in capturing discriminative spatiotemporal features for violence detection across four benchmark datasets: Hockey Fights, RLVS, Surveillance Fight, and AIRTLab, under optimized hyperparameters (batch size 8 or 16, learning rate 0.0001), with results consolidated in Table 4. All results were obtained using five-fold cross-validation, and the reported metrics represented the average across folds to ensure robustness and reliability.

It was observed from the table results that the proposed CNN-based model consistently achieved top-tier performance across all datasets and evaluation metrics. For instance, on Hockey Fights, it attained the highest accuracy (98.96%) and AUC (99.87%), along with superior recall (99.11%), precision (98.82%), and F1-score (98.96%) for the violent class, while also recording the lowest inference time (4.32 s). Moreover, it outperformed competing architectures by 1.75–2.24% in accuracy, 1.59–2.18% in recall, 1.68–2.26% in precision, and 1.73–2.22% in F1-score. In addition, it achieved a reduction in AUC of up to 0.35%, with testing time reduced by 15.6–55.2%. Similarly, on RLVS, it led with 98.54% accuracy and 99.75% AUC and achieved the lowest loss (0.050). In addition, it showed improvements of up to 1.7% in accuracy, 2.14% in precision, and 1.51% in F1-score over alternative backbones, while reducing testing time by up to 54.4%. Notably, only VGG16 achieves a marginally higher AUC by 0.02%.

However, the Surveillance Fight dataset presents a more challenging scenario. In this case, VGG16 achieves the highest accuracy, followed by

Table 4
Performance comparison of the proposed model and pretrained CNN-based variants across multiple datasets.

Dataset	Model	Category	Recall	Precision	F1-score	Loss	Acc.	AUC	Inference Time (S)
Hockey Fights B = 8, LR= 0.0001	CNN+ ConvLSTM + temporal attention	Violent	**99.11**	**98.82**	**98.96**	**0.041**	**98.96**	**99.87**	**4.32**
		Non-violent	**98.79**	**99.10**	**98.95**				
	VGG16+ ConvLSTM + temporal attention	Violent	97.52	96.77	97.14	0.075	97.11	99.70	8.46
		Non-violent	96.70	97.49	97.09				
	VGG19+ ConvLSTM + temporal attention	Violent	97.33	97.14	97.23	0.078	97.21	99.60	9.64
		Non-violent	97.09	97.28	97.19				
	MobileNetV2+ ConvLSTM + temporal attention	Violent	96.93	96.56	96.74	0.098	96.72	99.52	5.12
		Non-violent	96.51	97.90	96.70				
RLVS B = 16, LR= 0.0001	CNN+ ConvLSTM + temporal attention	Violent	98.76	**98.63**	**98.70**	**0.050**	**98.54**	99.75	**32.81**
		Non-violent	**98.28**	98.43	**98.34**				
	VGG16+ ConvLSTM + temporal attention	Violent	**99.03**	98.27	98.65	0.051	98.48	**99.77**	62.81
		Non-violent	97.79	**98.77**	98.27				
	VGG19+ ConvLSTM + temporal attention	Violent	98.96	98.22	98.59	0.050	98.42	99.70	72.00
		Non-violent	97.72	98.66	98.19				
	MobileNetV2+ ConvLSTM + temporal attention	Violent	97.91	96.49	97.19	0.090	96.84	99.37	40.36
		Non-violent	95.49	97.31	96.39				
Surveillance Fight B = 8, LR= 0.0001	CNN+ ConvLSTM + temporal attention	Violent	96.91	92.38	94.57	0.193	94.10	98.11	**2.32**
		Non-violent	90.93	96.34	93.53				
	VGG16+ ConvLSTM + temporal attention	Violent	98.14	93.02	**95.49**	**0.151**	**95.08**	98.30	4.15
		Non-violent	91.62	97.80	94.59				
	VGG19+ ConvLSTM + temporal attention	Violent	**98.76**	92.20	95.34	0.156	94.86	**98.55**	4.73
		Non-violent	90.46	**98.52**	94.28				
	MobileNetV2+ ConvLSTM + temporal attention	Violent	95.26	93.37	94.28	0.152	93.88	98.50	3.35
		Non-violent	92.34	94.59	93.42				
AIRTLab B = 8, LR= 0.0001	CNN+ ConvLSTM + temporal attention	Violent	**99.49**	**99.25**	**99.37**	**0.023**	**99.15**	**99.97**	**7.23**
		Non-violent	**98.45**	**98.974**	**98.704**				
	VGG16+ ConvLSTM + temporal attention	Violent	98.74	98.55	98.64	0.048	98.16	99.85	14.72
		Non-violent	97.00	97.43	97.18				
	VGG19+ ConvLSTM + temporal attention	Violent	98.87	98.29	98.57	0.057	98.08	99.81	16.77
		Non-violent	96.47	97.69	97.05				
	MobileNetV2+ ConvLSTM + temporal attention	Violent	98.02	98.08	98.05	0.084	97.37	99.42	8.22
		Non-violent	96.03	96.0	95.98				

VGG19, with the proposed model ranking third; nevertheless, the accuracy gap remains below 1%. Despite this, The CNN-based model still recorded the lowest testing time across all experiments. It also outperformed MobileNetV2 by 0.22% in accuracy, 1.65% in recall, and 0.29% in F1-score, while being 30.7% faster. The relatively lower performance on this dataset can be attributed to its unique challenges, including crowded scenes, varied backgrounds, low resolution, noise, and inconsistent lighting. These factors introduce subtle and ambiguous indicators of violence. Consequently, pre-trained VGG models, trained on large-scale datasets like ImageNet, are better suited to extract generalizable features under such conditions. As illustrated in Fig. 5, these challenges lead to misclassification by the CNN-based model, including both false negative and false positive cases. In contrast, the VGG-based model correctly classifies the same samples.

In contrast, on AIRTLab, the model achieved its strongest results: 99.15% accuracy, 99.97% AUC, and the lowest loss (0.023). Additionally, it gains of up to 1.78% in accuracy, 1.47% in recall, 1.17% in precision, and 1.32% in F1-score, alongside a testing time reduced by 12.0% to 56.9%.

Overall, the experimental results confirm the robustness and efficiency of the proposed CNN + ConvLSTM + Temporal Attention model. It achieves the best balance of classification accuracy, recall, precision, F1-score, AUC, and computational efficiency across the majority of evaluated datasets This establishing it as the most effective and practical architecture for real-time violence detection.

4.4.2. Quantitative and qualitative evaluation of the proposed model

This subsection evaluates the performance of the proposed CNN + ConvLSTM + Temporal Attention model across four benchmark datasets. The model is assessed using the best-performing configuration, as mentioned earlier. The evaluation includes both quantitative and qualitative analyses, such as ROC curves, confusion matrices, and temporal attention visualization, to provide a comprehensive understanding of the model's behavior.

- **ROC Curve Analysis**

To further illustrate the quantitative performance of the selected

Fig. 5. Misclassified samples by the CNN-based model and corresponding correct predictions by the VGG-based model for Surveillance Fight dataset: (a) CNN + ConvLSTM + Temporal Attention, (b) VGG16 + ConvLSTM + Temporal Attention.

best-performing configuration, Fig. 6 presents the cross-validation ROC curves of the proposed CNN + ConvLSTM + Temporal Attention model across four datasets. Each curve represents an individual fold, while the mean ROC curve and its standard deviation summarize the overall performance across the five folds. The model achieves near-perfect performance on the Hockey Fights and AIRTLab datasets, with mean AUC values of 0.9987 ± 0.0013 and 0.9997 ± 0.0002, respectively, demonstrating high accuracy and stability. Similarly, strong performance is observed on the RLVS dataset (AUC = 0.9975 ± 0.0004). In contrast, slightly lower performance is obtained on the Surveillance Fight dataset (AUC = 0.9811 ± 0.0105). This reflects the challenges of complex scenes, occlusions, and low-resolution footage.

- **Confusion Matrix Analysis**

Fig. 7 presents the cross-validation confusion matrices of the proposed model across four datasets. The model demonstrates highly accurate and consistent classification on the Hockey Fights and AIRTLab datasets, with minimal misclassifications, indicating stable performance across folds. Slightly higher error rates are observed on the RLVS and Surveillance Fight datasets. These results reflect the challenges of real-world conditions such as occlusion, motion blur, and low-resolution footage. This is also evident in the corresponding ROC results. Despite these challenges, the model maintains robust detection performance across all datasets.

- **Analysis of Temporal Attention Weights**

For additional performance evaluation of the proposed model, attention weights are visualized across 16 consecutive frames for both violent and non-violent samples from the hockey fight dataset, as shown in Fig. 8. The attention weights are normalized using a softmax function, ensuring that their sum equals one across all frames. It can be observed that the weights are relatively evenly distributed, with only slight variations between frames. This shows the model successfully captures long-range temporal dependencies rather than over-relying on isolated, on a single “trigger” frame. In the violent sample Fig. 8(a), the attention weights show slight variations, there is a gradual increase in weight from F1 to F5 This indicates the frames where physical interaction between players become more pronounced and visually complex. In contrast, the

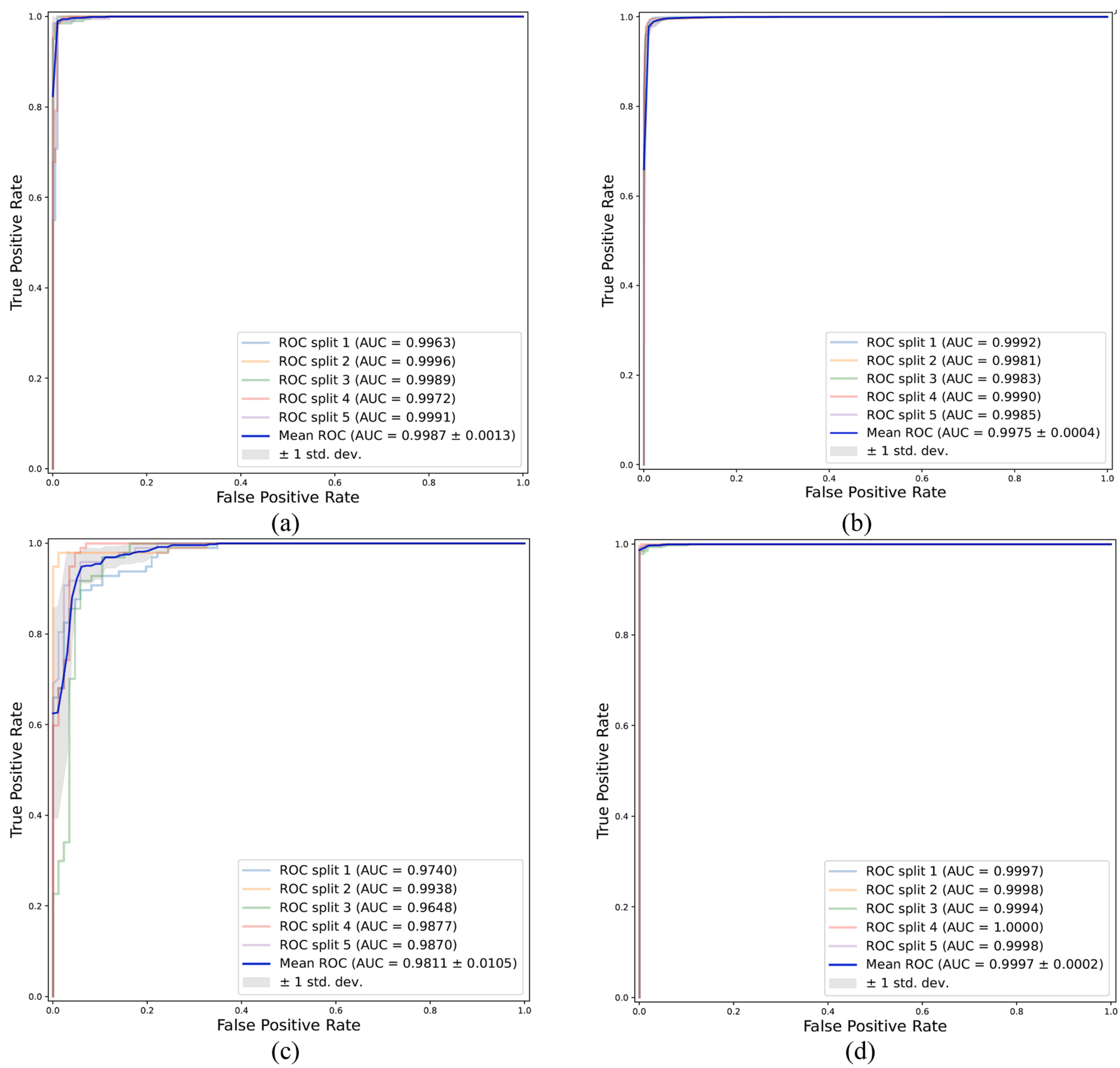

Fig. 6. Cross-validation ROC curves of the CNN + ConvLSTM + Attention model on the (a) Hockey Fights dataset. (b) RLVS dataset. (c) Surveillance Fight dataset. (d) AIRTLab dataset.

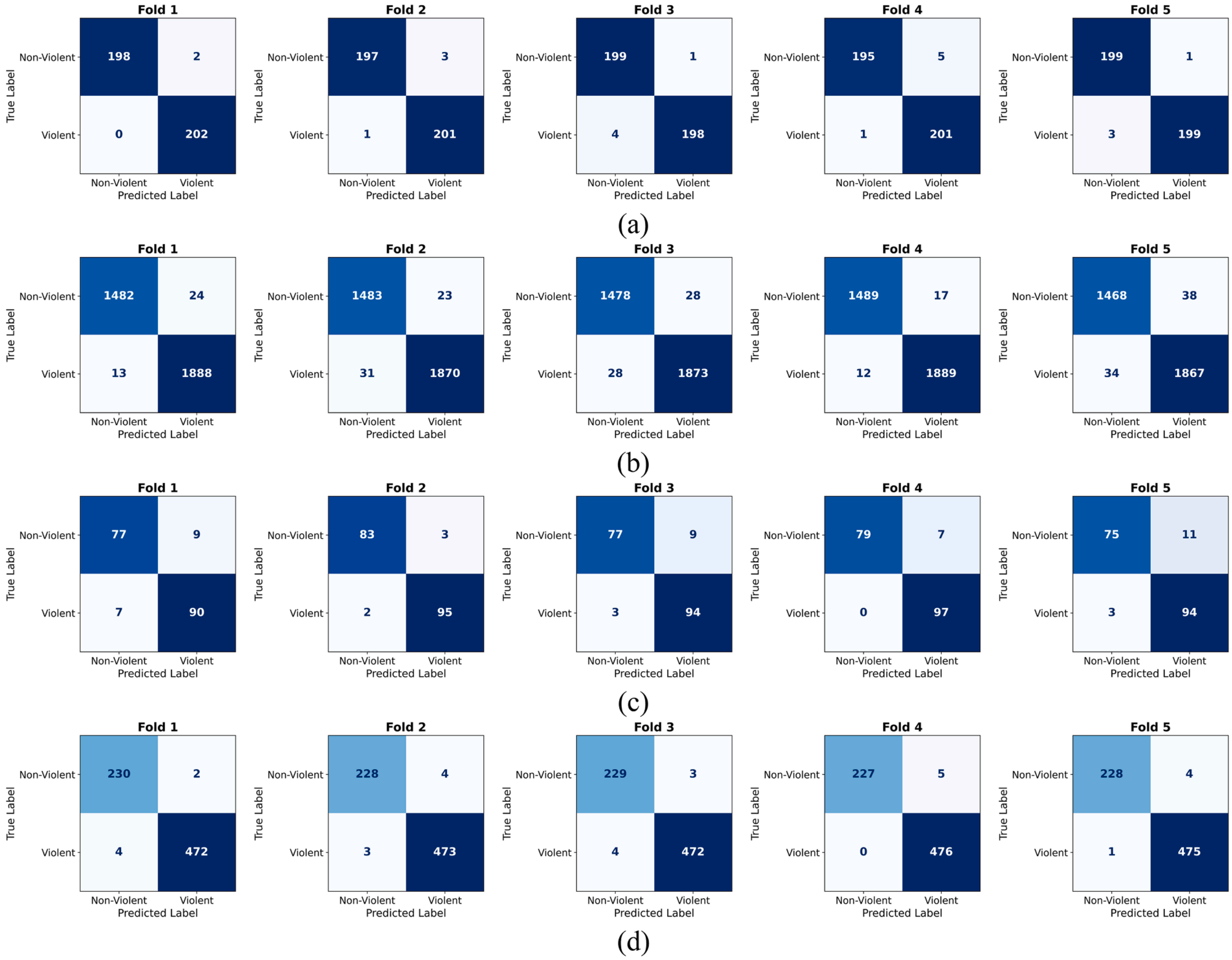

Fig. 7. Confusion matrices across five-fold cross-validation of the CNN + ConvLSTM + Attention model on the (a) Hockey Fights dataset. (b) RLVS dataset. (c) Surveillance fight dataset (d) AIRTLab dataset.

non-violent sample Fig. 8(b) exhibits a more uniform distribution with minimal variations. This reflects the stable and continuous nature of the scene. This difference suggests that the model assigns greater importance to frames containing more distinctive motion patterns in violent scenarios, while distributing attention more evenly when no significant activity is present.

• **Qualitative Analysis of Model Predictions**

To further illustrate the qualitative performance of the proposed model, Fig. 9 presents representative prediction samples across four benchmark datasets. The samples highlight correct classifications of both violent and non-violent scenarios under diverse and challenging conditions, including occlusion, camera motion, and varying viewpoints. These examples demonstrate the model's ability to capture discriminative spatiotemporal patterns, even in visually ambiguous cases. While some misclassifications are observed in more challenging datasets such as RLVS and Surveillance Fight, as reflected in the confusion matrix and ROC analyses, the model overall exhibits strong robustness and generalization capability across different surveillance environments.

4.4.3. Comparative analysis with conventional violence detection approaches

In this section, the performance of the proposed hybrid model is compared with eighteen conventional violence detection methods published between 2021 and 2025, using the previously described datasets. Table 5 presents the results of this comparative study, highlighting the performance of the proposed CNN combined with ConvLSTM and temporal attention against conventional methods across the evaluated datasets. Results marked with "–" indicate that the corresponding metric was not reported in the original publications. All comparative results are directly sourced from the respective authors' papers.

4.4.3.1. Datasets-based results. • **Performance on Hockey Fights Dataset**

The proposed model achieved 98.96% accuracy and F1-score, with an AUC of 99.87%. This represents a 0.46% accuracy improvement over the closest competitors, MobileNetV3 + Bi-LSTM + Bi-LTMA [37] and ST-TCN + Bottleneck Attention + Bottleneck Transformer [19] (both 98.50%), and 3.0% over the lowest-performing competitor, ConvLSTM + Residual Attention [35] (95.99%). The F1-score is 0.04% below the highest value of 99.00% achieved by EfficientNetB0 + Bi-LTMA + TSM [15], and 2.96% above the lowest of 96% [35]. The AUC is 0.87% higher than the lowest competitor (99.00% [35]) and 0.06% below the highest (99.93% [37]).

• **Performance on RLVS Dataset**

The model achieved 98.54% accuracy, 98.70% F1-score, and 99.75% AUC. Accuracy improvements range from 0.24% over Mobile Video Network A0 [33] (98.30%) to 12.0% over the lowest-performing competitor, Two Conv3D Layers [9] (88.00%). The F1-score surpasses the next-best method by 0.69% (Features Fusion + KNN [45]: 98.01%) and exceeds the lowest-performing competitor by 5.5% (MobileNetV3 +

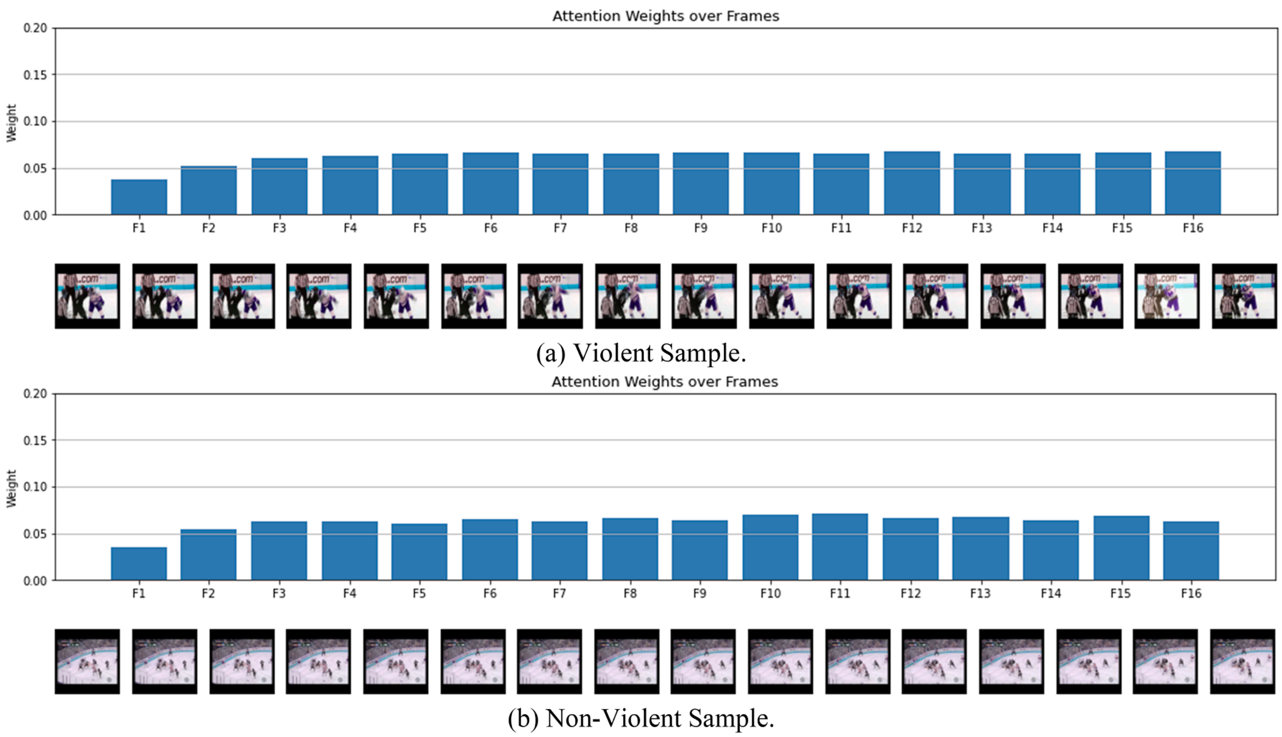

Fig. 8. Visualization of temporal attention weights across video frames for violent and non-violent samples from hockey fights dataset.

Bi-LSTM + Bi-LTMA [37]: 93.2%). The AUC improves upon X3D-M [1] (99.60%) by 0.15%.

• **Performance on Surveillance Fight Dataset**

The proposed model achieved an accuracy of 94.10% and an F1-score of 94.57%. These results reflect improvements ranging from approximately 1.60% over the closest competitor, ST-TCN + Bottleneck Attention + Bottleneck Transformer [19] (92.5%), to 30.7% over the lowest-performing method, Xception + BiLSTM + Self-attention [44] (72%), in accuracy. For F1-score, the proposed model was 0.43% lower than the next best method, EfficientNetB0 combined with Bi-LTMA and TSM [15], which achieved an F1-score of 95.00%.

• **Performance on AIRTLab Dataset**

The model achieved 99.15% accuracy, 99.37% F1-score, and 99.97% AUC. Accuracy improvements span from 0.86% over Conv2D + ConvLSTM + Conv2D [31] (98.29%) to 9.15% over the lowest-performing competitor, ConvLSTM + Residual Attention [35] (90%). F1-score improvements range from 1.37% over PCA + LR [4] (98%) to 9.37% over the lowest-performing competitor [35] (90%). The AUC exceeds VGG16 + ConvLSTM [8] (99.11%) by 0.86% and outperforms the lowest-performing competitor [35] (97%) by 2.97%.

4.4.3.2. Complexity-performance analysis. To evaluate the proposed model's trade-off between complexity and performance, its results are first compared with lower-complexity methods. Traditional approaches, such as handcrafted spatial-motion descriptors [34] and PCA combined with logistic regression [4], as well as lightweight models like MobileNetV2 with ConvLSTM [29], present limited capability in capturing temporal dynamics. These methods underperform by 0.76% to 5.05% in accuracy compared to the proposed CNN-based model. This highlights their limitations in modeling the complex spatiotemporal patterns found in surveillance video.

As model complexity increases, moderately deep architectures such as Mobile Video Network A0 [33], Conv2D+ ConvLSTM+ Conv2D [31], Five Conv+ GRU+ Dense [26], U-Net (MobileNetV2)+ LSTM [25], VGG16+ ConvLSTM [8], ConvLSTM + Residual Attention [35], and Two Conv3D Layers [9] introduce additional convolutional and temporal modeling capacity. However, despite their increased complexity compared to the proposed model, the proposed approach outperforms them, achieving accuracy improvements ranging from approximately 0.24% to 10.54%. This underscores the effectiveness of the proposed model in capturing complex spatiotemporal dynamics in surveillance video.

At a higher complexity level, architectures such as Xception+ LSTM [28], Xception+ BiLSTM + Self attention [44], Feature Fusion+ LogReg, [7], and Feature Fusion+ KNN [45] incorporate deeper feature extractors and sequence models. Despite their representational power, the proposed method maintains a competitive edge, showing accuracy gains ranging from 0.55% up to 22.1%. This is largely due to its more efficient integration of temporal attention and ConvLSTM layers.

At the top end of complexity, models such as EfficientNetB0+ Bi-LTMA+ TSM [15], Ensemble of (ResNet50 or InceptionResnetv2 or DenseNet121+LSTM) [10], MobileNetV3 + Bi-LSTM + Bi-LTMA [37], and X3D-M [1] offer high accuracy but at the cost of computational efficiency. Among these, EfficientNetB0 + Bi-LTMA + TSM [15] slightly outperformed the proposed model by 0.43% in accuracy on the Surveillance Fight dataset. In contrast, the remaining models exhibited performance deficits ranging from 0.46% to 5.29% in accuracy. Despite their deep architecture and temporal modeling capabilities. These methods typically require significantly more training and inference resources. This may limit their suitability for real-time or resource-constrained surveillance applications.

In conclusion, despite being relatively lightweight, the proposed CNN-based hybrid model not only achieves the highest overall accuracy and F1-score mostly across all datasets but also maintains better runtime efficiency than many deeper or ensemble-based models. The performance improvements range from 0.24% to 30.7% in accuracy and up to 32.5% in F1-score. This emphasizes the advantage of carefully designed mid-level architectures that balance spatial-temporal representation with computational feasibility.

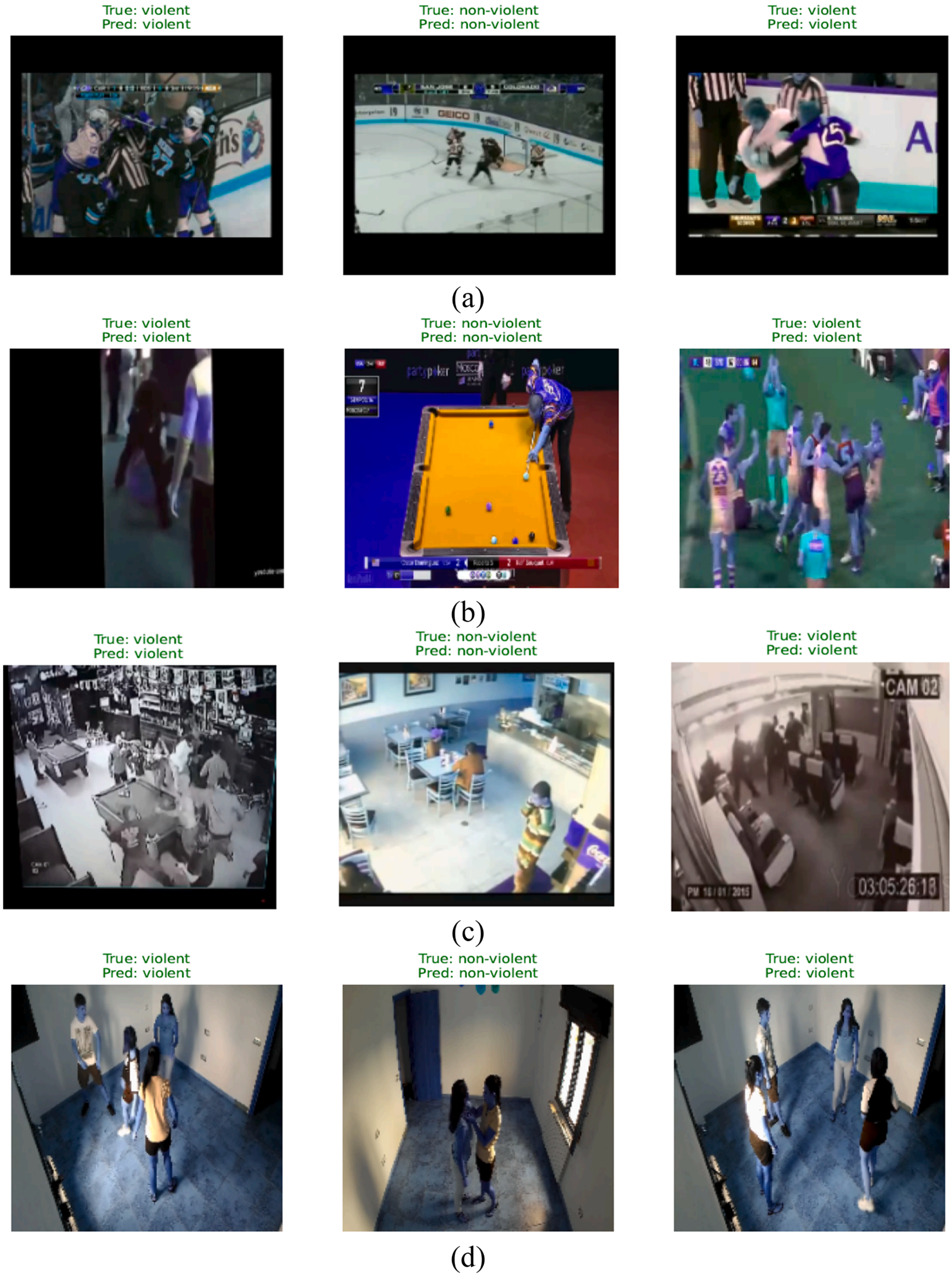

Fig. 9. Prediction Samples of the CNN + ConvLSTM + Attention model on the (a) Hockey Fights dataset. (b) RLVS dataset. (c) Surveillance Fight dataset (d) AIRTLab dataset.

Table 5

Comparison of the proposed hybrid model with other methods on the hockey fights, RLVS, surveillance fight, and AIRTLab datasets.

Model	Hockey Fights			RLVS			Surveillance Fight			AIRTLab		
	Acc.	F1	AUC	Acc.	F1	AUC	Acc.	F1	AUC	Acc.	F1	AUC
Xception+ BiLSTM + Self attention [44] 2019	–	–	–	–	–	–	72	–	–	–	–	–
Xception+ LSTM [28] 2021	96.55	–	–	–	–	–	–	–	–	–	–	–
VGG16+ ConvLSTM [8] 2021	97.31	97.34	99.63	–	–	–	–	–	–	95.62	96.77	99.11
U-Net (MobileNetV2)+ LSTM [25] 2022	96.1	0.961	–	–	–	–	–	–	–	–	–	–
Con2D+ ConvLSTM+ Conv2D [31] 2022	–	–	–	–	–	–	–	–	–	98.29	97.66	-
MobileNetV2+ConvLSTM [29] 2023	–	–	–	–	–	–	–	–	–	94.10	95.62	98.26
ConvLSTM + Residual Attention [35] 2023	95.99	96	99	–	–	–	–	–	–	90	90	97
PCA + LR [4] 2023										98.31	98	98
X3D-M [1], 2023	–	–	–	96.5	–	99.6	–	–	–	–	–	–
Features Fusion (Xception, InceptionV3, InceptionResNetV2)+ LogReg, [7] 2023	–	–	–	97.66	97.67	–	–	–	–	–	–	–
Spatial Motion Extractor+ Short & Global Temporal Extractor [34] 2024	98.2	–	–	–	–	–	–	–	–	–	–	–
EfficientNetB0 + Bi-LTMA+ TSM [15] 2024	98.5	**99**	–	–	–	–	91.67	**95**	–	–	–	–
Five 2D Conv+ GRU+ 6 Dense [26] 2024	97.62	–	–	–	–	–	–	–	–	97.22	–	–
Two conv3D layers [9], 2024	–	–	–	88	–	–	–	–	–	–	–	–
Ensemble of (ResNet50 or InceptionResnetv2 or DenseNet121+LSTM) [10], 2024	–	–	–	96.6	96.6	–	–	–	–	–	–	–
ST-TCN + Bottleneck Attention + Bottleneck Transformer [19], 2024	98.5	98.0	98.5	–	–	–	92.5	92.0	92.5	–	–	–
Features Fusion (MobileNetV2, InceptionV3, InceptionResNetV2, and Xception)+ KNN [45], 2025	–	–	–	97.99	98.01	–	–	–	–	–	–	–
Mobile Video Network A0 [33] 2025	98.0	–	–	98.3	–	–	84	–	–	–	–	–
MobileNetV3+ Bi-LSTM+ Bi-LTMA [37] 2025	98.50	98.49	**99.93**	93.25	93.20	98.17	–	–	–	–	–	–
Proposed Method	**98.96**	98.96	99.87	**98.54**	**98.70**	**99.75**	**94.10**	94.57	**98.11**	**99.15**	**99.37**	**99.97**

4.5. Ablation study

This section evaluates the impact of different design choices within the proposed model. Specifically, it investigates the effect of varying the CNN architecture, including the number of convolutional blocks and filter sizes, as well as the contribution of the ConvLSTM and Temporal Attention modules. The analysis aims to provide a comprehensive understanding of how each component influences the overall performance of the model.

4.5.1. Impact of CNN architectural design

This subsection analyzes the impact of different CNN architectural configurations on the performance of the proposed model. In particular, the number of convolutional blocks and the number of filters is varied to examine their influence on feature representation and overall detection performance. The model is evaluated using configurations with five and six convolutional blocks. In these configurations, the number of filters increases progressively (16–512 for six blocks, and 16–256 or 32–512 for five blocks).

The performance of each configuration is assessed across all datasets using recall, precision, F1-score, accuracy, AUC, loss, and inference time, as summarized in Table 6. The results indicate that the configuration with six convolutional blocks consistently achieves the best overall performance. In this configuration, where the number of filters

Table 6

Performance of the proposed CNN + ConvLSTM + Temporal attention model under different configurations with varying filter sizes and numbers of convolutional blocks.

Dataset	CNN Configuration (Filters, Blocks)	Category	Recall	Precision	F1-score	Loss	Acc.	AUC	Inference Time (S)
Hockey Fights	**(16–512, 6 blocks)**	Violent	**99.11**	**98.82**	**98.96**	**0.041**	**98.96**	**99.87**	**4.32**
		Non-violent	**98.79**	**99.10**	**98.95**				
	(16–256, 5 blocks)	Violent	98.91	98.33	98.62	0.045	98.61	99.89	4.84
		Non-violent	98.31	98.91	98.60				
	32–256, 5 blocks)	Violent	98.71	99.01	98.86	0.037	98.86	99.89	4.63
		Non-violent	99.01	98.71	98.86				
RLVS	**(16–512, 6 blocks)**	**Violent**	**98.76**	**98.63**	**98.70**	0.051	98.54	99.75	32.81
		Non-violent	**98.28**	**98.43**	**98.34**				
	(16–256, 5 blocks)	Violent	97.80	97.93	97.86	0.068	97.62	99.62	36.14
		Non-violent	97.38	97.23	97.30				
	32–256, 5 blocks)	Violent	98.12	97.89	98.00	0.065	97.77	99.69	36.40
		Non-violent	97.32	97.63	97.48				
Surveillance Fight	**(16–512, 6 blocks)**	**Violent**	**96.91**	**92.38**	**94.57**	**0.193**	**94.10**	**98.11**	**2.32**
		Non-violent	**90.93**	**96.34**	**93.53**				
	(16–256, 5 blocks)	Violent	96.49	89.91	93.07	0.196	92.35	97.69	2.37
		Non-violent	87.67	95.65	91.47				
	32–256, 5 blocks)	Violent	96.08	91.73	93.85	0.193	93.33	97.61	2.40
		Non-violent	90.24	95.35	92.73				
AIRTLab	**(16–512, 6 blocks)**	**Violent**	**99.49**	**99.246**	**99.37**	**0.023**	**99.15**	**99.97**	**7.23**
		Non-violent	**98.45**	**98.974**	**98.704**				
	(16–256, 5 blocks)	Violent	99.37	98.79	99.08	0.036	98.76	99.92	7.76
		Non-violent	97.49	98.71	98.10				
	32–256, 5 blocks)	Violent	99.45	99.29	99.37	0.028	**99.15**	99.95	7.80
		Non-violent	98.53	98.87	98.71				

increases from 16 to 512 across layers, the model attains higher accuracy, F1-score, and AUC, while maintaining low loss and competitive inference time. In contrast, configurations with five convolutional blocks exhibit slightly lower performance across most evaluation metrics, despite having comparable computational cost. This trend is particularly noticeable on the Surveillance Fight dataset due to its higher complexity. This further highlighting the advantage of the deeper six-block configuration. Therefore, the selected best-performing six-block architecture is consistently used across all previous experiments to ensure a fair and unified evaluation.

4.5.2. Impact of temporal modeling and attention mechanisms

This subsection presents an ablation study to evaluate the impact of key architectural components, specifically the choice of temporal modeling unit (LSTM vs. ConvLSTM) and the inclusion of a temporal attention mechanism. Table 7 summarizes the performance of four model configurations across all datasets: CNN + LSTM, CNN + ConvLSTM, CNN + LSTM + Temporal Attention, and CNN + ConvLSTM + Temporal Attention.

- **Effect of Temporal Modeling (ConvLSTM vs. LSTM):**

Replacing LSTM with the proposed ConvLSTM had a mixed but generally modest effect overall. On Hockey Fights, both CNN + LSTM and CNN + ConvLSTM achieved identical accuracy of 98.51% and nearly identical F1-scores of 98.52% for both classes. This suggests that on relatively constrained and controlled footage, the spatial memory capacity of ConvLSTM offers no immediate advantage over standard LSTM without the support of temporal attention. On RLVS, CNN + ConvLSTM recorded a slightly lower accuracy of 97.13% versus 97.71% for CNN + LSTM, and a reduced AUC of 99.49% versus 99.64%, indicating that without temporal attention, ConvLSTM's added spatial complexity may introduce optimization difficulties on more diverse video data. A similar pattern was observed on the Surveillance Fight dataset, where CNN + ConvLSTM achieved 88.85% accuracy compared to 89.62% for CNN + LSTM, though both remained well below the attention-augmented variants. On AIRTLab, the two configurations were closely matched, with CNN + ConvLSTM achieving 98.79% accuracy versus 98.56% for CNN + LSTM, a marginal improvement of 0.23%.

- Effect of Temporal Attention

Temporal attention consistently improved performance across both configurations. The gains were considerably larger when applied to the proposed ConvLSTM. On Hockey Fights, accuracy improved from 98.51% to 98.86% for the LSTM variant and from 98.51% to 98.96% for the proposed model. AUC rose from 99.81% to 99.88% and from 99.75% to 99.87% respectively. The proposed model also further reduced inference time from 4.52 s to 4.32 s. On RLVS, the LSTM variant achieved a modest accuracy gain from 97.71% to 97.76%, with loss reduced from 0.074 to 0.071. The proposed model showed a substantially larger improvement. Accuracy rose from 97.13% to 99.15%, AUC from 99.49% to 99.97%, and loss dropped from 0.084 to 0.023. On Surveillance Fight, the LSTM variant improved from 89.62% to 91.48% in accuracy and from 90.67% to 92.09% in violent class F1-score. The proposed model achieved larger gains, with accuracy rising from 88.85% to 94.10%, AUC from 94.97% to 98.11%, and violent class F1-score from 89.59% to 94.57%. This represents the largest absolute gain observed across all configuration transitions in this dataset. On AIRTLab, the LSTM variant improved from 98.56% to 98.98% in accuracy and from 99.91% to 99.96% in AUC. The proposed model reached peak performance of 99.15% accuracy and 99.97% AUC, recording the lowest loss of 0.023. The violent class F1-score also improved from 99.10% to 99.37%.

Overall, these results confirm that ConvLSTM alone does not consistently outperform LSTM. However, temporal attention is the decisive factor that unlocks the full potential of the proposed architecture. The CNN + ConvLSTM + Temporal Attention model achieves the best performance across nearly all metrics and datasets, confirming that spatiotemporal feature preservation combined with selective temporal weighting is the most effective strategy for violence detection.

Table 7
Evaluation of temporal modeling approaches (ConvLSTM vs. LSTM) and the effect of temporal attention across different datasets.

Dataset	Model	Category	Recall	Precision	F1-score	Loss	Acc.	AUC	Inference Time (S)
Hockey Fights	CNN + LSTM	Violent	99.01	98.05	98.52	0.060	98.51	99.81	4.79
		Non-violent	98.01	98.99	98.50				
	CNN + ConvLSTM	Violent	98.61	98.43	98.52	0.064	98.51	99.75	4.52
		Non-violent	98.41	98.59	98.50				
	CNN + LSTM + **Temporal Attention**	Violent	98.61	99.11	98.86	0.044	98.86	99.88	4.72
		Non-violent	99.11	98.61	98.86				
	CNN+ ConvLSTM + Temporal Attention	Violent	**99.11**	**98.82**	**98.96**	**0.041**	**98.96**	**99.87**	**4.32**
		Non-violent	**98.79**	**99.10**	**98.95**				
RLVS	CNN + LSTM	Violent	97.92	97.99	97.96	0.074	97.71	99.64	36.12
		Non-violent	97.46	97.39	97.42				
	CNN + ConvLSTM	Violent	97.61	97.26	97.43	0.084	97.13	99.49	36.18
		Non-violent	96.63	96.88	96.75				
	CNN + LSTM + **Temporal Attention**	Violent	97.99	98.00	98.00	0.071	97.76	99.64	35.36
		Non-violent	97.47	97.48	97.46				
	CNN+ ConvLSTM + Temporal Attention	**Violent**	**98.76**	**98.63**	**98.70**	**0.051**	**98.54**	**99.75**	**32.81**
		Non-violent	**98.28**	**98.43**	**98.34**				
Surveillance Fight	CNN + LSTM	Violent	95.05	86.76	90.67	0.286	89.62	95.97	2.46
		Non-violent	83.49	93.82	88.27				
	CNN + ConvLSTM	Violent	90.52	88.85	89.59	0.281	88.85	94.97	**2.32**
		Non-violent	86.98	89.25	87.97				
	CNN + LSTM + **Temporal Attention**	Violent	94.02	90.39	92.09	0.224	91.48	96.80	2.48
		Non-violent	88.60	93.17	90.73				
	CNN+ ConvLSTM + Temporal Attention	**Violent**	**96.91**	**92.38**	**94.57**	**0.193**	**94.10**	**98.11**	**2.32**
		Non-violent	**90.93**	**96.34**	**93.53**				
AIRTLab	CNN + LSTM	Violent	99.16	98.71	98.93	0.037	98.56	99.91	7.76
		Non-violent	97.32	98.25	97.79				
	CNN + ConvLSTM	Violent	99.16	99.04	99.10	0.035	98.79	99.93	7.76
		Non-violent	98.02	98.28	98.14				
	CNN + LSTM + **Temporal Attention**	Violent	99.12	99.37	99.24	0.030	98.98	99.96	8.15
		Non-violent	98.70	98.21	98.46				
	CNN+ ConvLSTM + Temporal Attention	**Violent**	**99.49**	**99.246**	**99.37**	**0.023**	**99.15**	**99.97**	**7.23**
		Non-violent	**98.45**	**98.974**	**98.70**				

5. Conclusion

In this work, a hybrid deep learning framework has been developed for violence detection in surveillance videos by integrating spatial, temporal, and attention-based representations. The proposed architecture combines a custom-designed CNN for feature extraction with ConvLSTM for spatiotemporal modeling and a temporal attention mechanism to emphasize critical motion patterns, enabling effective understanding of complex video dynamics. Experimental results across four benchmark datasets demonstrate that the proposed model achieves strong and consistent performance, reaching accuracies of up to 99.15% while reducing inference time by up to 56.9%. Compared to existing approaches, the model provides a favorable balance between accuracy and efficiency, making it suitable for real-time applications. Although slightly lower performance is observed on the Surveillance Fight dataset, the model maintains a minimal accuracy gap while offering significantly faster processing. Overall, the findings highlight the effectiveness of combining ConvLSTM with temporal attention within a lightweight architecture for practical violence detection. However, the model is currently evaluated using dataset-specific training, which may limit its generalization to unseen domains.

Future work will focus on enhancing robustness through cross-dataset evaluation and domain adaptation techniques to better handle domain shifts. Additionally, further optimization will be explored to support deployment on edge devices and resource-constrained environments. In addition, extending the framework to support early violence detection from partial video sequences could enable faster response in real-time scenarios. Furthermore, integrating multimodal information and enhancing performance under challenging conditions such as occlusion and low resolution will be explored.

Declaration of generative AI in scientific writing

During the preparation of this manuscript, the authors used AI tools for the purposes of improving grammar, language clarity, and structure, with no substantive contribution to the originality, ideas, or scientific content of the work. After using these tools, the authors reviewed and revised the content as needed and take full responsibility for the scientific integrity and originality of the manuscript.

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CRediT authorship contribution statement

Azza El-Fiky: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Nawal El-Fishawy:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Athar A. Ein-Shoka:** Writing – review & editing, Supervision. **Randa Mohamed Bayoumi:** Writing – review & editing, Validation, Supervision, Software, Methodology, Conceptualization. **Rasha Shoitan:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available and cited within the article.

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